MIT **Fall 2026**
Department of Economics **14.461 (first half)**

Advanced Macroeconomics I: Identification in Macroeconomics

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Lectures:	Mon/Wed 10:30am–12:00pm (Room E52-432)
Recitations:	Fri 1:00pm-2:30pm (Room E51-390)
Office hours:	Instructor: by appointment (Room E52-520)
	TA: by appointment
Website:	https://canvas.mit.edu/courses/39004

Description. This is a class in identification in macroeconomics, covering empirical methods and their relation to structural macroeconomic modeling, with a focus on business cycles. The key overarching objective is to teach students how to combine data and modeling structure to think about the classical questions of business-cycle analysis, from the origins of business cycles to the efficacy and desirability of stabilization policy.

Prerequisites. Completion of the macroeconomics and econometrics classes in the first-year PhD sequence. Students from outside the Economics PhD program should contact the instructor to obtain permission to take the course.

Material. The class will be largely organized around required readings. Attached to this syllabus is a list of relevant papers; the starred (*) papers are essential reading before the lecture, while the optional readings are useful for a deeper understanding of the material. Handouts for each lecture will be made available on the course website.

While there is no textbook for the course, some of the time series material is sufficiently standard so that good textbook treatments are available. Good reference books include:

Brockwell, P. J., and Davis, R. A. (1991). *Time Series: Theory and Methods*. 2nd edition, Springer. (Very nice mathematical treatment of the classic theory of covariance stationary time series, though not aimed at economists.)

Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. (Comprehensive reference for time series econometrics methods developed before the mid-1990s.)

Cochrane, J. H. (2005). *Time series for macroeconomics and finance*. Manuscript, University of Chicago. (Accessible introduction to linear time series methods.)

Kilian, L., and Lütkepohl, H. (2017). *Structural Vector Autoregressive Analysis*. Cambridge University Press. (Recent reference on SVAR methods.)

Parts of my lecture notes build on a class taught by Mikkel Plagborg-Møller.

Evaluation. Your performance will be evaluated based on class participation (30%), a problem set (30%), and a referee report plus replication of a paper (40%).

1. *Class participation.* Starred (*) papers are essential reading before the lecture. Parts of each class will revolve around discussion of those papers, so it is important that you are equipped to critically assess them.

2. *Problem set.* The problem set will be posted at the beginning of the semester, and covers material related to all parts of the course. You are encouraged to complete the relevant exercises as we progress through the material.

3. *Referee report & paper replication.* For the referee report, a background document for refereeing best practice is posted on the course webpage. The paper should be unpublished; beyond that you can freely pick it according to your interests, subject to instructor approval. The replication exercise is for the same paper, and involves both replicating the main findings and then providing one small extension (e.g., a different sample period, a change in the empirical specification, a twist in the model, ...).

Reporting bias-related or other incidents. The Economics Department fully endorses the Institute Discrimination and Harassment Response Office (IDHR) reporting process for any bias-related incident.[1] We encourage students to use this valuable resource if needed for any reason. This process is useful for addressing any issues that may arise with professors, teaching assistants, or other students in this course. If you have had a negative experience and are not sure if IDHR is the place to go, they can also help you find the right office at MIT to receive support. The department values such reports as important to our pursuit of equitable and inclusive treatment for all students, faculty, and staff.

[1] You can submit a report at `https://idhr.mit.edu/submitincidentreport`.

Accessibility and support. The Economics Department values an inclusive environment. If you need a disability accommodation to access this course, please communicate with us early in the semester. If you have your accommodation letter, please meet with the faculty so that we can understand your needs and implement approved accommodations. If you have not yet been approved for accommodations, please contact Disability and Access Services at `das-student@mit.edu` to learn about their procedures. We encourage you to do so early in the term to allow sufficient time for implementation of accommodations that you may need.

AI guidance. If you think it will be helpful, then you should feel free to explore the use of artificial intelligence (AI) tools for the problem set, referee report, and paper replication. Any such use must be appropriately acknowledged and cited. It is each student's responsibility to assess the validity of any AI output that is submitted; you bear the final responsibility. Violations of this policy will be considered academic misconduct.

Important dates. Below is a (preliminary) summary of important dates for this class. Changes will be announced via course email.

September 9 (Wed): First class

October 9 (Fri): Referee report due

October 13 (Tue): Monday schedule on Tuesday

October 16 (Fri): Paper replication due

October 21 (Wed): Last class

October 23 (Fri): Problem set due

Course outline

1. Overview: the impulse-propagation framework *(≈ Lecture 1)*

    - Background: What is a macroeconomic "shock"? Why does macroeconomics take a shock-propagation perspective? How do we think about policy counterfactuals?
    - Outline for the rest of the class

2. Refresher: linear models *(≈ Lectures 2-3)*

    a) Linearized structural macroeconomic models

        - State-space methods
        - Sequence-space methods
        - Structural vector moving average (SVMA) representation

    b) Linear time series methods

        - Lag operators, linear filters, VARMA, spectral analysis
        - Wold decomposition

3. Policy and policy causality in the impulse-propagation paradigm *(≈ Lectures 4-6)*

    - History of thought: Cowles Commission, Lucas critique, Sims program
    - Shock space vs. instrument space, policy shocks vs. policy rules in modern macroeconomic models
    - Equilibrium selection and policy regime change

4. Empirical time series methods for shock identification & estimation *(≈ Lectures 7-8)*

    a) Identification assumptions

        - Background: the SVMA model identification challenge
        - Classical solutions: invertibility + exclusion restrictions (short-run, long-run), sign/magnitude restrictions, heteroskedasticity, . . . ; use of instruments/proxies
        - Recent alternative: leverage time variation in the structure of the economy

    b) Estimation strategies

        - Overview: VAR, LP (+ intermediate shrinkage techniques)
        - LP/VAR population equivalence & finite-sample recommendations

c) Non-linear time series methods *(if time permits)*

- General potential outcomes framework
- Interpretation of non-linear LPs

5. Applications *(≈ Lectures 9-10)*

a) "What-if" scenarios for policy

- Role of empirics vs. model structure in policy counterfactual evaluation
- Effects of more aggressive interest rate hikes post-2021

b) Origins of business-cycle fluctuations

- "Main business-cycle shock" (business-cycle anatomy), model estimation through full-information methods
- TFP shocks vs. investment-specific technology shocks, business-cycle decomposition via NK-DSGEs (incl. HANK)

c) Identifying structural macroeconomic equations

- Regressions in impulse response space
- Application to Phillips curve estimation

6. Cross-sectional analysis *(≈ Lectures 11-12)*

a) What does cross-sectional variation identify?

- Cross-household/firm analysis, channel decompositions
- Cross-regional analysis & local general equilibrium effects

b) The "missing intercept"/aggregation problem

- "Micro"-based aggregation
- Leveraging shock equivalence

c) Decomposition of shocks and policy transmission into channels

Reading list

Essential readings are listed first and marked with a star (*). Other readings are included for your reference. Original contributions are not always cited when good handbook/textbook references are available.

1 Overview: macroeconomic shocks

* Ramey, V. A. (2016). "Macroeconomic Shocks and Their Propagation." In *Handbook of Macroeconomics, Volume 2A*, edited by Taylor, J. B., and Uhlig, H., Elsevier, 71–162.

* Nakamura, E., and Steinsson, J. (2018). "Identification in Macroeconomics." *Journal of Economic Perspectives* 32(3), 59–86.

Summers, L. H. (1991). "The Scientific Illusion in Empirical Macroeconomics." *The Scandinavian Journal of Economics* 93(2), 129–148.

Beaudry, P., Galizia, D., and Portier, F. (2020). "Putting the Cycle Back into Business Cycle Analysis." *American Economic Review*, 110(1), 1–47.

2 Refresher: linear models

Linearized structural macro models

* Fernández-Villaverde, J., Rubio-Ramírez, J. F., and Schorfheide, F. (2016). "Solution and estimation methods for DSGE models." In *Handbook of Macroeconomics, Volume 2A*, edited by Taylor, J. B., and Uhlig, H., Elsevier, 527–724.

* Auclert, A., Bardóczy, B., Rognlie, M., and Straub, L. (2021). "Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models." *Econometrica*, 89(5), 2375–2408.

Fernández-Villaverde, J., Rubio-Ramírez, J. F., Sargent, T. J., and Watson, M. W. (2007). "ABCs (and Ds) of Understanding VARs." *American Economic Review* 97(3), 1021–1026.

Smets, F., and Wouters, R. (2007). "Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach." *American Economic Review*, 97(3), 586–606.

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (2005). "Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy." *Journal of Political Economy*, 113(1), 1–45.

Boppart, T., Krusell, P., and Mitman, K. (2018). "Exploiting MIT Shocks in Heterogeneous-Agent Economies: the Impulse Response as a Numerical Derivative." *Journal of Economic Dynamics and Control*, 89, 68–92.

Linear time series models

* Hamilton: chapters 2–4, 10–12.

* Kilian and Lütkepohl: chapters 2.1–2.5.

Brockwell and Davis: chapters 1.1–1.5, 2.1–2.9, 3.1–3.5, 5.1–5.5, 5.7, 11.1–11.4.

Cochrane: chapters 3–6, 8.

3 Policy in the impulse-propagation paradigm

History of thought: "Lucas program" and Sims-Zha approach

* Christiano, L. J., Eichenbaum, M., and Evans, C. L. (1999). "Monetary Policy Shocks: What Have We Learned and to What End?" *Handbook of Macroeconomics*, 1, 65–148.

* Bernanke, B. S., Gertler, M., and Watson, M. (1997). "Systematic Monetary Policy and the Effects of Oil Price Shocks." *Brookings Papers on Economic Activity*, 1997(1), 91–157.

Sims, C. A., and Zha, T. (2006). "Does Monetary Policy Generate Recessions?" *Macroeconomic Dynamics*, 10(2), 231–272.

Hamilton, J. D., and Herrera, A. M. (2004). "Oil Shocks and Aggregate Macroeconomic Behavior: the Role of Monetary Policy: Comment." *Journal of Money, Credit and Banking*, 36(2), 265–286.

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (2005). "Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy." *Journal of Political Economy*, 113(1), 1–45.

Chari, V. V., Kehoe, P. J., and McGrattan, E. R. (2009). "New Keynesian Models: Not Yet Useful for Policy Analysis." *American Economic Journal: Macroeconomics*, 1(1), 242–266.

Kilian, L., and Lewis, L. T. (2011). "Does the Fed Respond to Oil Price Shocks?" *The Economic Journal*, 121(555), 1047–1072.

Antolín-Díaz, J., Petrella, I., and Rubio-Ramírez, J. F. (2021). "Structural Scenario Analysis with SVARs." *Journal of Monetary Economics*, 117, 798–815.

Rules vs. shocks in modern structural models

* McKay, A., and Wolf, C. K. (2023). "What Can Time-Series Regressions Tell Us About Policy Counterfactuals?" *Econometrica*, 91(5), 1695–1725.

* Caravello, T. E., McKay, A., and Wolf, C. K. (2026). "Evaluating Policy Counterfactuals: When Do We Need Structural Models?" Working Paper.

Barnichon, R., and Mesters, G. (2023). "A Sufficient Statistics Approach for Macro Policy." *American Economic Review*, 113(11), 2809–2845.

4 Time series methods for shock identification and estimation

Identification assumptions

1. *Identification problem, invertibility*

* Fernández-Villaverde, J., Rubio-Ramírez, J. F., Sargent, T. J., and Watson, M. W. (2007). "ABCs (and Ds) of Understanding VARs." *American Economic Review* 97(3), 1021–1026.

Lippi, M., and Reichlin, L. (1994). "VAR Analysis, Nonfundamental Representations, Blaschke matrices." *Journal of Econometrics* 63(1), 307–325.

Leeper, E. M., Walker, T. B., and Yang, S. C. S. (2013). "Fiscal Foresight and Information Flows." *Econometrica*, 81(3), 1115–1145.

Forni, M., Gambetti, L., and Sala, L. (2019). "Structural VARs and Noninvertible Macroeconomic Models." *Journal of Applied Econometrics* 34(2), 221–246.

Plagborg-Møller, M., and Wolf, C. K. (2022). "Instrumental Variable Identification of Dynamic Variance Decompositions." *Journal of Political Economy*, 130(8), 2164–2202.

2. *Exclusion restrictions*

* Christiano, L. J., Eichenbaum, M., and Evans, C. L. (1999). "Monetary Policy Shocks: What Have We Learned and to What End?" *Handbook of Macroeconomics*, 1, 65–148.

* Blanchard, O., and Quah, D. (1989). "The Dynamic Effects of Aggregate Demand and Supply Disturbances." *American Economic Review*, 79(4), 655–673.

Blanchard, O., and Perotti, R. (2002). "An Empirical Characterization of the Dynamic Effects of Changes in Government Spending and Taxes on Output." *The Quarterly Journal of Economics*, 117(4), 1329–1368.

Ramey, V. A. (2011). "Identifying Government Spending Shocks: It's All in the Timing." *The Quarterly Journal of Economics*, 126(1), 1–50.

3. *Sign/magnitude restrictions*

* Uhlig, H. (2005). "What are the Effects of Monetary Policy on Output? Results from an agnostic identification procedure." *Journal of Monetary Economics* 52(2), 381–419.

* Baumeister, C., and Hamilton, J. D. (2015). "Sign Restrictions, Structural Vector Autoregressions, and Useful Prior Information." *Econometrica* 83(5), 1963–1999.

Mountford, A., and Uhlig, H. (2009). "What are the Effects of Fiscal Policy Shocks?" *Journal of Applied Econometrics*, 24(6), 960–992.

Giacomini, R., and Kitagawa, T. (2021). "Robust Bayesian Inference for Set-Identified Models." *Econometrica*, 89(4), 1519–1556.

Wolf, C. K. (2022). "What Can We Learn from Sign-Restricted VARs?" *AEA Papers and Proceedings* 112, 471–75.

4. *Other invertibility-based methods*

* Montiel Olea, J. L., Plagborg-Møller, M., and Qian, E. (2022). "SVAR Identification From Higher Moments: Has the Simultaneous Causality Problem Been Solved?" *AEA Papers and Proceedings* 112, 481–485.

* Kolesár, M., and Plagborg-Møller, M. (2025). "Dynamic Causal Effects in a Nonlinear World: the Good, the Bad, and the Ugly." *Journal of Business & Economic Statistics*, 43(4), 737–754.

Rigobon, R. (2003). "Identification Through Heteroskedasticity." *Review of Economics and Statistics* 85(4), 777–792.

Gouriéroux, C., Monfort, A., and Renne, J.-P. (2017). "Statistical Inference for Independent Component Analysis: Application to Structural VAR Models." *Journal of Econometrics* 196(1), 111–126.

Brunnermeier, M., Palia, D., Sastry, K. A., and Sims, C. A. (2021). "Feedbacks: Financial Markets and Economic Activity." *American Economic Review*, 111(6), 1845–79.

5. *Instruments/proxies*

* Stock, J. H., and Watson, M. W. (2018). "Identification and Estimation of Dynamic Causal Effects in Macroeconomics Using External Instruments." *Economic Journal* 128(610), 917–948.

* Plagborg-Møller, M., and Wolf, C. K. (2022). "Instrumental Variable Identification of Dynamic Variance Decompositions." *Journal of Political Economy*, 130(8), 2164–2202.

Gertler, M., and Karadi, P. (2015). "Monetary Policy Surprises, Credit Costs, and Economic Activity." *American Economic Journal: Macroeconomics* 7(1), 44–76.

Nakamura, E., and Steinsson, J. (2018). "High-Frequency Identification of Monetary Non-Neutrality: the Information Effect." *The Quarterly Journal of Economics*, 133(3), 1283–1330.

Känzig, D. R. (2021). "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements." *American Economic Review*, 111(4), 1092–1125.

6. *Time variation in policy*

* Manuel, E., and Wolf, C. K. (2026). "Identifying Policy Causal Effects from Rule Changes." Working Paper.

Hack, L., Istrefi, K., and Meier, M. (2024). "Identification of Systematic Monetary Policy." Working Paper.

Estimation strategies

* Rubio-Ramírez, J. F., Waggoner, D. F., and Zha, T. (2010). "Structural Vector Autoregressions: Theory of Identification and Algorithms for Inference." *Review of Economic Studies* 77(2), 665–696.

* Plagborg-Møller, M., and Wolf, C. K. (2021). "Local Projections and VARs Estimate the Same Impulse Responses." *Econometrica*, 89(2), 955–980.

* Montiel Olea, J. L., Plagborg-Møller, M., Qian, E., and Wolf, C. K. (2025). "Local Projections or VARs? A Primer for Macroeconomists." *NBER Macroeconomics Annual*, 40.

Sims, C. A. (1980). "Macroeconomics and Reality." *Econometrica* 48(1), 1–48.

Jordà, O. (2005). "Estimation and Inference of Impulse Responses by Local Projections." *American Economic Review* 95(1), 161–182.

Barnichon, R., and Brownlees, C. (2019). "Impulse Response Estimation By Smooth Local Projections." *Review of Economics and Statistics* 101(3), 522–530.

Montiel Olea, J. L., and Plagborg-Møller, M. (2021). "Local Projection Inference is Simpler and More Robust Than You Think." *Econometrica*, 89(4), 1789–1823.

Li, D., Plagborg-Møller, M., and Wolf, C. K. (2024). "Local Projections vs. VARs: Lessons From Thousands of DGPs." *Journal of Econometrics*, 244(2).

Montiel Olea, J. L., Plagborg-Møller, M., Qian, E., and Wolf, C. K. (2026). "Double Robustness of Local Projections and Some Unpleasant VARithmetic." *Econometrica*, 94(4), 1313–1343.

Non-linear time series methods

* Rambachan, A., and Shephard, N. (2021). "When Do Common Time Series Estimands have Nonparametric Causal Meaning?" Working Paper.

* Kolesár, M., and Plagborg-Møller, M. (2025). "Dynamic Causal Effects in a Nonlinear World: the Good, the Bad, and the Ugly." *Journal of Business & Economic Statistics*, 43(4), 737–754.

Gonçalves, S., Herrera, A. M., Kilian, L., and Pesavento, E. (2024). "State-Dependent Local Projections." *Journal of Econometrics*, 244(2).

Caravello, T. E., and Martínez-Bruera, P. (2024). "Disentangling Sign and Size Non-Linearities." Working Paper.

5 Applications

"What-if" scenarios for policy

* Caravello, T. E., McKay, A., and Wolf, C. K. (2026). "Evaluating Policy Counterfactuals: When Do We Need Structural Models?" Working Paper.

Barnichon, R. (2022). "What If? Monetary Policy in Hindsight." *FRBSF Economic Letter*, 2022-28.

Barnichon, R., and Mesters, G. (2023). "A Sufficient Statistics Approach for Macro Policy." *American Economic Review*, 113(11), 2809–2845.

Wolf, C. K. (2023). "Fiscal Stimulus and the Systematic Response of Monetary Policy." *AEA Papers and Proceedings*, 113, 388–393.

Origins of business-cycle fluctuations

* Ramey, V. A. (2016). "Macroeconomic Shocks and Their Propagation." In *Handbook of Macroeconomics, Volume 2A*, edited by Taylor, J. B., and Uhlig, H., Elsevier, 71–162.

* Angeletos, G. M., Collard, F., and Dellas, H. (2020). "Business-Cycle Anatomy." *American Economic Review*, 110(10), 3030–70.

Justiniano, A., Primiceri, G. E., and Tambalotti, A. (2010). "Investment Shocks and Business Cycles." *Journal of Monetary Economics*, 57(2), 132–145.

Auclert, A., Rognlie, M., and Straub, L. (2020). "Micro Jumps, Macro Humps: Monetary Policy and Business Cycles in an Estimated HANK Model." NBER Working Paper.

Identifying structural macroeconomic equations

* Barnichon, R., and Mesters, G. (2020). "Identifying Modern Macro Equations with Old Shocks." *The Quarterly Journal of Economics*, 135(4), 2255–2298.

* Liu, L., Plagborg-Møller, M., and Tan, N. M. P. (2026). "Limited-Information Estimation of Heterogeneous Agent Models." arXiv preprint arXiv:2608.13953.

Galí, J., and Gertler, M. (1999). "Inflation Dynamics: A Structural Econometric Analysis." *Journal of Monetary Economics*, 44(2), 195–222.

Campbell, J. Y., and Mankiw, N. G. (1989). "Consumption, Income, and Interest Rates: Reinterpreting the Time Series Evidence." *NBER Macroeconomics Annual*, 4, 185–216.

Mavroeidis, S., Plagborg-Møller, M., and Stock, J. H. (2014). "Empirical Evidence on Inflation Expectations in the New Keynesian Phillips Curve." *Journal of Economic Literature*, 52(1), 124–88.

Hazell, J., Herreño, J., Nakamura, E., and Steinsson, J. (2022). "The Slope of the Phillips Curve: Evidence from US States." *The Quarterly Journal of Economics*, 137(3), 1299–1344.

Lewis, D. J., and Mertens, K. (2023). "Dynamic Identification Using System Projections and Instrumental Variables." Working Paper.

6 Cross-sectional analysis

What does cross-sectional variation identify?

* Nakamura, E., and Steinsson, J. (2014). "Fiscal Stimulus in a Monetary Union: Evidence from US Regions." *American Economic Review*, 104(3), 753–92.

* Auclert, A., and Rognlie, M. (2018). "Inequality and Aggregate Demand." Working Paper.

Koby, Y., and Wolf, C. K. (2020). "Aggregation in Heterogeneous-Firm Models: Theory and Measurement." Working Paper.

Wolf, C. K. (2025). "Interest Rate Cuts versus Stimulus Payments: An Equivalence Result." *Journal of Political Economy*, 133(4), 1235–1275.

The "missing intercept"/aggregation problem

* Chodorow-Reich, G. (2019). "Geographic Cross-Sectional Fiscal Spending Multipliers: What Have We Learned?" *American Economic Journal: Economic Policy*, 11(2), 1–34.

* Wolf, C. K. (2023). "The Missing Intercept: A Demand Equivalence Approach." *American Economic Review*, 113(8), 2232–2269.

Guren, A., McKay, A., Nakamura, E., and Steinsson, J. (2021). "What Do We Learn from Cross-Regional Empirical Estimates in Macroeconomics?" *NBER Macroeconomics Annual*, 35(1), 175–223.

Chodorow-Reich, G., Nenov, P. T., and Simsek, A. (2021). "Stock Market Wealth and the Real Economy: A Local Labor Market Approach." *American Economic Review*, 111(5), 1613–57.

Channel decompositions

* Holm, M. B., Paul, P., and Tischbirek, A. (2021). "The Transmission of Monetary Policy under the Microscope." *Journal of Political Economy*, 129(10), 2861–2904.

Kaplan, G., Moll, B., and Violante, G. L. (2018). "Monetary Policy According to HANK." *American Economic Review*, 108(3), 697–743.